

Relating individual differences in internalizing symptoms to emotional attention set-shifting in children

Oana Mocan^a, Oana Stanciu^b and Laura Visu-Petra^{a*}

^a*Department of Psychology, Developmental Psychology Lab, Babeş-Bolyai University, Cluj-Napoca, Romania;* ^b*School of Philosophy, Psychology and Social Sciences, The University of Edinburgh, Edinburgh, UK*

(Received 10 September 2012; accepted 25 November 2013)

A growing body of evidence points to links between internalizing symptoms and various executive functioning deficits, and especially to inhibition and set-shifting difficulties. However, there is limited developmental research regarding the impact of internalizing symptoms on the shifting function, particularly during middle childhood. The current study investigated attention shifting in a sample of 108 early school age children (7–11 years) using a task-switching paradigm which required participants to alternate between emotional and nonemotional judgments. Results indicated that higher levels of internalizing symptoms (anxiety and depression) had a detrimental effect on performance efficiency (measured by response times) but not on response accuracy. This effect was only observed on emotional (and not on nonemotional) repetition trials and did not affect switching trials; moreover, it was only present when feedback was presented to participants. The findings partially support the predictions of the Attentional Control Theory in a developmental sample and suggest that individual differences in internalizing symptoms play a role in children's ability to flexibly alternate between emotional judgments.

Keywords: internalizing; anxiety; depression; emotional attention set-shifting; attentional control; children

Introduction

The ever-changing environment requires us to flexibly adjust our thoughts and actions to daily life requirements. This flexibility is supported by complex cognitive control processes which allow us to produce meaningful, goal-directed behavior, and to adapt this behavior according to situational demands (e.g. Botvinick, Braver, Barch, Carter, & Cohen, 2001). The current study focuses on set-shifting, a function deployed when switching back and forth between multiple tasks, operations, or mental sets (Monsell, 1996). The task-switching paradigm is an influential framework developed to investigate set-shifting (e.g. Allport, Styles, & Hsieh, 1994; Rogers & Monsell, 1995). During a typical task-switching test, participants have to alternate between making judgments about one of two dimensions of the same stimulus, for instance, its color or shape, on a trial-by-trial basis.

There is a substantial body of research regarding task switching in adults (reviewed in Kiesel et al., 2010; Monsell, 2003); yet, little is known about the developmental trajectory of performance on this task, particularly beyond the preschool years (Cragg & Nation,

*Corresponding author. Email: laurapetra@psychology.ro

2009). Fine-grained developmental studies targeting middle childhood pointed to significant improvements in the ability to set-shift during this interval (Davidson, Amso, Cruess Anderson, & Diamond, 2006; Huizinga, Dolan, & van der Molen, 2006) and indicated a reduction in interference from an irrelevant task (Cragg & Nation, 2009). This progress could be supported by substantial improvements in attentional control during this period (Heim & Keil, 2012).

Research on dispositional factors that generate individual differences in set-shifting across the life span is even more limited. In adults, individual differences in *anxiety* have been shown to impact shifting ability. The Attentional Control Theory (ACT; Eysenck, Derakshan, Santos, & Calvo, 2007) proposes that higher levels of anxiety primarily impair attentional control, which leads to performance decrements in executive functions such as inhibition or shifting (Derakshan & Eysenck, 2009). Several studies on adult populations corroborate the predictions of the ACT (Ansari, Derakshan, & Richards, 2008; Derakshan, Smyth, & Eysenck, 2009). Depressive symptoms have also been linked to performance impairments in adult populations (i.e. more perseverative errors) on classical set-shifting tasks such as the Wisconsin Card Sorting Test (see Rogers et al., 2004, for a review). Given the high comorbidity of anxiety and depression symptoms especially in childhood and adolescence (Lonigan, Carey, & Finch, 1994), the use of a composite index of internalizing symptoms might be best suited for this developmental period. Early internalizing problems have been linked to the inability to set-shift in a clinical versus a nonclinical sample of boys aged 9–11 (Emerson, Mollet, & Harrison, 2005). On the other hand, in a preschool nonclinical sample, high-anxious (HA) girls displayed better set-shifting performance than low-anxious (LA) girls, but presented longer response times (RTs; Tincas, Dragos, Ionescu, & Benga, 2007).

An important prediction of the ACT is that anxious individuals will exhibit executive deficits particularly in disengaging attention from *emotional* stimuli (Eysenck et al., 2007). Therefore, in task-switching contexts, HA individuals are expected to show an increased emotional to neutral switch cost, and a reduced neutral to emotional switch cost compared to LA individuals. In contrast, the Cognitive Avoidance Theory (Sibrava & Borkovec, 2006) predicts that HA individuals avoid attending to emotionally charged stimuli. Consequently, they are expected to manifest an increased neutral to emotional switch cost and a reduced emotional to neutral switch cost when compared to LA individuals. Consistent with this latter prediction, Johnson (2009a) found that individuals high in anxiety and worry exhibited a deficit in the ability to switch from a neutral to an emotional mental set (this was also observed in the case of individuals with high levels of depressive symptoms and tendencies to ruminate, De Lissnyder, Koster, Derakshan, & De Raedt, 2010).

In the developmental population, the role of attentional control as a moderator for the relationship between internalizing symptoms and set-shifting has been under-investigated. In adults, skilled control of voluntary attention may allow anxious people to limit the impact of threatening information (Derryberry & Reed, 2002). Across development, attentional control may have an important function in buffering negative emotions by helping children reduce their emotional reactivity when confronted with stressful or challenging events (Perez-Edgar & Fox, 2007).

The main theoretical and empirical goal of our study was to relate individual differences in internalizing symptoms (anxiety and depression) to children's ability to

deploy attention toward and away from emotional stimuli during middle childhood. To be able to assess emotional attentional control during middle childhood, an initial methodological aim was to adapt the attentional control capacity for emotion (ACCE) task used by Johnson (2009a, 2009b) in a study with adults. The use of this task in a younger sample could yield important insights into the development of internally directed emotional attentional control during this period. Since we aimed to investigate the effects of stimulus valence, we maintained the demand for participants to alternate between emotional and nonemotional judgments, while the switch itself became sensitive to the emotional valence of the stimuli (as compared to the always neutral switches in the previously mentioned study). Second, we wanted to check whether the potentially negative effect of internalizing symptoms on set-shifting performance was moderated by self-perceived attentional control capacity. Finally, we wanted to explore the effect of externally directed performance feedback, compared to a non-feedback condition, and relate this effect to individual differences in internalizing symptoms.

Following the reviewed literature and the abovementioned goals, it became clear that our investigation was essentially exploratory. First, in order to confirm the validity of the ACCE task for developmental populations, we expected to replicate the findings of the literature on task switching in adults (as reviewed by Monsell, 2003): an overall poorer performance (accuracy and RTs) for task switching compared to task repetition trials and shorter RTs at longer cue-to-stimulus intervals (CSIs), compared to short CSIs. From a developmental point of view, we expected that with age, performance on this task would become faster and more accurate.

According to the predictions of the ACT (Eysenck et al., 2007), individual differences in internalizing symptoms (especially in trait anxiety) were expected to have a greater detrimental effect on processing efficiency (measured by RTs) than on performance effectiveness (measured by accuracy). Furthermore, as the adverse effects of trait anxiety were thought to be exacerbated by an increase in central executive demands, we speculated that performance on switching trials would suffer more interference from internalizing symptoms than performance on repetition trials. Based on Johnson's (2009b) findings, we expected anxious children to experience difficulties (longer RTs) in switching from a neutral to an emotional set, supporting the Cognitive Avoidance Theory of anxiety (Sibrava & Borkovec, 2006). An alternative possibility, suggested by the ACT, would be that anxious children would experience greater difficulty in switching from an emotional to a neutral set, reflecting difficulties in disengaging an emotional set. Since the investigation of the impact of valence and trial-by-trial feedback in the ACCE task was exploratory, no specific hypotheses could be advanced.

Method

Participants

A sample of 108 primary school children (54 girls) aged between 7 and 11 ($M = 9.23$ years, $SD = 1.16$) were recruited from a local Romanian school. The children came from families which varied widely in household earnings and maternal education levels (45% of mothers had completed at least an undergraduate degree, and an additional 31% had graduated from high school).

Procedure

Written consent was obtained from parents, and children gave their verbal assent to take part in the study. The children who were old enough to read and write ($n = 86$) completed two self-report questionnaires measuring attentional control (the Child version of the Attentional Control Scale [ACS-C]) and internalizing symptoms (the Revised Child Anxiety and Depression Scale [RCADS]) and all parents filled in the parental report version of the same scale (RCADS-P). Following this, all children were administered the computerized experimental task individually under the guidance of the experimenter, completing the feedback and non-feedback conditions in counterbalanced order (in two distinct sessions which took place a few days apart).

Materials

The RCADS (Chorpita, Yim, Moffitt, Umemoto, & Francis, 2000) translated into Romanian by Tincas, Cheie, Mocan, Benga, and Visu-Petra (in preparation) is a 47-item self-report scale adapted from the Spence Children's Anxiety Scale (Spence, 1997, 1998). RCADS is used for measuring the frequency of occurrence of different anxiety (37 items) and depression symptoms (10 items). Items are scored from 0 to 3 corresponding to "never," "sometimes," "often," and "always." The presence of symptoms was assessed through both child and parent reports.

The ACCE task, adapted from Johnson (2009a), uses an explicit task-cuing paradigm to measure an individual's ability to internally shift attention between emotional and neutral mental sets. In the current experiment, children had to make two types of judgments on a compound stimulus that consisted of faces located within geometrical shapes (see Figure 1). A cue presented either 200 ms (short CSI) or 1500 ms (long CSI) before stimulus onset indicated whether a neutral or emotional judgment had to be made. Following stimulus presentation, participants had to press one of three keys (1, 2, or 3) according to whether they thought this was a happy, neutral, or angry face (emotion judgment), or a circle, triangle, or square (shape judgment).

Johnson (2009a) wanted to minimize the reaction of participants to emotional faces during switching and, therefore, the valence of the stimuli was always neutral for switching trials. We manipulated the valence of the stimuli, on both repetition and switching trials. This resulted in five types of switching trials as a function of the emotionality of the stimuli: positive to neutral, neutral to positive, negative to neutral, neutral to negative, and neutral to neutral. It is important to note that there were four more types of repetition trials according to valence, thus making the design unbalanced.

Each participant completed two conditions of the task, one with and one without trial-by-trial feedback. When feedback was provided, response accuracy was indicated by a sign displayed for 1500 ms following each response (a green check sign for a correct answer and a red $\times$ sign for a wrong answer). If no response was provided within a 5000 ms period from stimulus onset, an hourglass was presented on the screen (and the software recorded the RTs as missing).

At the beginning of the ACCE task, children received a short period of training to familiarize themselves with the stimuli used in the experiment and learn the stimulus-response mappings. This consisted of 3 practice blocks of 25 trials each corresponding to emotional (facial expression), neutral (shape type), and mixed (emotional and neutral) judgments. A sticker illustrating the mappings was visible at all times allowing the

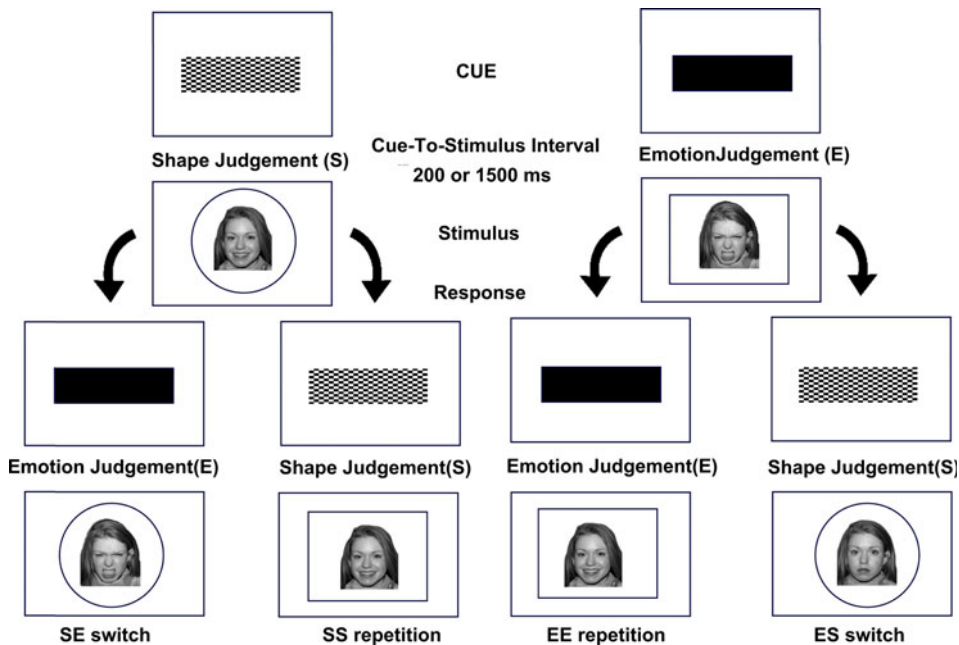


Figure 1. A schematic representation of the ACCE task. Each trial began with the presentation of a cue which informed the child which judgment to make: an emotional judgment (E) or a shape judgment (S). The cue remained on the screen for 200 ms or 1500 ms until the stimulus was presented and the participants had to make a judgment by pressing a key. Afterward, a blank screen was presented for an interval of 500 ms followed by the next trial. Thus, the trial could entail either a task switch (emotional to shape judgment or shape to emotional judgment) or a task repetition (emotion to emotion judgment or shape to shape judgment).

Note: SE switch = shape-emotion switching trials; SS repetition = shape-shape repetition trials; EE repetition = emotion-emotion repetition trials; ES switch = emotion-shape switching trials.

children to consult it when in doubt (thus diminishing the memory load of the task). Halfway into the experimental task, the children were given a short break in order to avoid fatigue.

A pseudorandomized list was created from 30 shape–emotion (SE) switching trials, 30 emotion–shape (ES) switching trials, 90 emotion–emotion (EE) repetition trials, and 30 shape–shape (SS) repetition trials. A larger number of EE trials were included in order to place a greater demand on attentional control for disengaging an emotional task set (Johnson, 2009a). The number of trials was identical in the feedback and non-feedback conditions (total of 360 trials).

Another constraint was applied, namely the first 10 test trials were emotional judgments (as in Johnson, 2009a, 2009b). The emotional stimuli consisted of photographs (one female and one male) from the MacBrain Face Stimulus Set (Tottenham, Borscheid, Ellertsen, Marcus, & Nelson, 2002). The experimental task was created and presented using the E-Prime 2.0 software.

The ACS-C (Derryberry & Reed, 2002) translated into Romanian by Susa, Pitica, Benga, and Miclea, (2012) is a 20-item self-report questionnaire which measures the ability to focus and shift attention. Items (e.g. “When concentrating, I do not notice what

happens around me”, “I can easily write or read, while I am talking on the phone”) are scored on a 4-point scale with 0 = never, 1 = sometimes, 2 = often, and 3 = always. Higher scores represent lower levels of attentional control.

Results

Questionnaire measures

The depression and anxiety subscales of the RCADS were highly correlated, child report: $r(84) = .71$, $p = .001$, two-tailed; parental report: $r(106) = .69$, $p = .001$, two-tailed. This convergence justified the use of the total score (internalizing) in subsequent analyses. The correlations between the child self-report and the parental report were moderate: $r(84) = .43$, $p = .001$, two-tailed, for the anxiety score, and $r(84) = .26$, $p = .01$, two-tailed, for the depression score. The child–parent correlation for the internalizing score was $r(84) = .42$, $p = .001$, two-tailed.^{1,2} Table 1 summarizes the scores obtained in the current study on both RCADS and ACS-C questionnaires together with gender specific means and standard deviations. We failed to uncover any age or gender-related differences in either the trait anxiety or the depression subscales of the RCADS. The following analysis will use the total RCADS (internalizing) score computed from the parental report.

We found no correlation between ACS scores and the children’s age, $r(84) = .11$, $p = .31$, two-tailed, or any effect of gender, $t(84) = .36$, $p = .7$, two-tailed. Surprisingly, we also found no significant relation between the child-reported ACS score and the parental internalizing score, $r(84) = .19$, $p = .072$, two-tailed.

Data analysis

The use of mixed linear models was preferred over general linear models as it allowed the direct modeling of RTs from all trials. This circumvents the use of the mean (or median) RTs within each condition as repeated outcome variables, which is necessary for analyses where switch costs are computed for each individual. The benefit lies in the ability to model unbalanced data as well as account for the codependency of observations. In particular, RTs are not independent as trials are nested within participants, and moreover, nearby trials are expected to be more similar than trials further apart (Field & Wright, 2011).

Separate models were built for RTs and accuracy data using the *nlme* (Pinheiro, Bates, DebRoy, Sarkar, & R Development Core Team, 2010) and *lme4* (Bates & Maechler, 2010) R packages. Following the guidelines provided in Field and Wright (2011), model comparisons were conducted through likelihood ratio tests for nested models and through the use of the Bayesian Information Criteria and Akaike Information Criteria otherwise. The *multcomp* package (Hothorn, Bretz, & Westfall, 2008) facilitated the correction for multiple comparisons (Tukey adjustment) and the *effects* package allowed the visualization of the effects (Fox, 2003).

Preliminary data analysis

Prior to the data analysis, the first trial of each experimental session was deleted. Additionally, 3.2% of trials were identified as outliers (RTs shorter than 200 ms) and were

Table 1. Descriptives for the questionnaire measures.

Measure	<i>M</i> (SD)					
	Parent report			Child report		
	Girls (<i>n</i> = 54)	Boys (<i>n</i> = 54)	Total (<i>n</i> = 108)	Girls (<i>n</i> = 42)	Boys (<i>n</i> = 44)	Total (<i>n</i> = 86)
RCADS anxiety	20.07 (12.97)	16.89 (9.59)	18.48 (11.47)	17.64 (11.90)	15.52 (11.53)	16.56 (11.69)
Depression	4.76 (2.86)	4.26 (2.76)	4.51 (2.80)	4.62 (3.62)	4.93 (3.68)	4.78 (3.63)
Total	24.83 (15.04)	21.15 (11.75)	22.99 (13.56)	22.26 (14.77)	20.45 (14.36)	21.34 (14.50)
ACS-C focusing				15.60 (3.88)	14.16 (4.17)	14.87 (4.07)
Shifting				10.07 (3.18)	10.84 (3.58)	10.47 (3.40)
Total				25.67 (5.83)	25.20 (6.11)	25.43 (5.94)

Note: RCADS = Revised Child Anxiety and Depression Scale; ACS-C = Attention Control Scale–Child version.

discarded. Trials on which participants provided a wrong answer (9.3%) or a correct answer following an erroneous response (4.2%) were eliminated from the analysis of RTs as it would be impossible to tell whether a task repetition or switch had been performed (Demanet, Liefoghe, & Verbruggen, 2011).

Overall, there were relatively few incorrect answers, on average 33.94 (SD = 16.66) per participant. As accuracy was not the main focus of the study and it was of interest only for disproving the existence of an speed-accuracy trade-off, we do not report the full model on accuracy here (the authors can be contacted for further details). Similar to the effects found in the model for RTs (see next section), we identified a higher probability of errors for switching trials, shorter CSIs, in older children, and an absence of gender, ACS or internalizing effects. These results can be observed graphically by plotting the estimated probabilities of failure as a function of trial type and age (see Figure 2). There was a minor decrease in the probability of an error when feedback was provided, which may be indicative of a feedback related speed-accuracy trade-off.

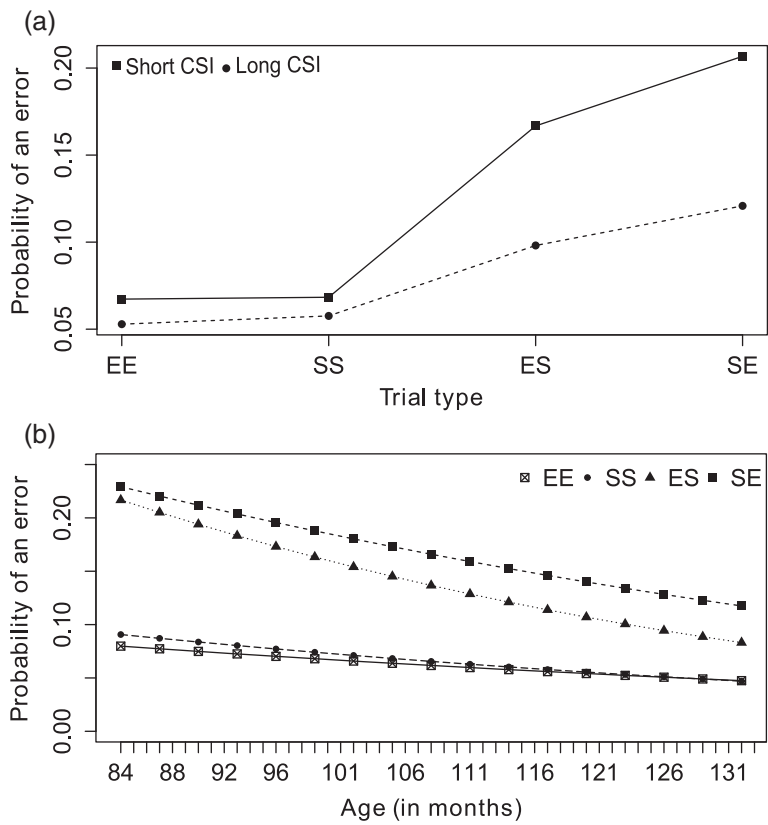


Figure 2. (a) Estimated probability of making an error by trial type and cue-to-stimulus interval (CSI). (b) Estimated probability of making an error as a function of age and trial type. Note: Short CSI = 200 ms; long CSI = 1500 ms; EE = emotion–emotion repetition; SS = shape–shape repetition; ES = emotion–shape switch; SE = shape–emotion switch.

Results

As the distribution of the RTs was heavily left skewed, a natural logarithmic transformation was applied in order to achieve a more normal distribution (Field & Wright, 2011). The reported model was fitted by restricted maximum likelihood estimation and a continuous autoregressive covariance structure was used over trials (within the two sessions). The attentional control score (from the ACS-C) did not prove to be a significant predictor and was dropped from the final model.

First, we present the random parameters in the model. As expected, there was a significant variation in the intercepts of the participants, $\text{var}(u_{0j}) = 0.04$, $LR = 4583$, $p < .001$. The slope of feedback also varied across participants, $\text{var}(u_{1j}) = 0.02$, $LR = 1676$, $p = .01$. Second, all of the experimental manipulations (the presence of feedback, trial type, CSI) were associated with significant changes in (geometric) mean RTs when controlling for the age, sex, and the internalizing scores of the participants as can be seen in Table 2a.

EE trials were chosen as the baseline because these were the most numerous trials. In what follows, we will discuss these predictors individually.

Effect of experimental manipulations

Overall, the estimated average log RTs differed across the four types of trials, $F(3, 32,201) = 36.77$, $p < .001$. Switching trials tended to elicit longer RTs than repetition trials as can be seen in Figure 3(a) which presents the estimated geometric means for the four types of trials. There was no significant difference in the estimated means of log RTs in the two types of repetition trials ($p = .15$), conditional on the other predictors. On the other hand, all else being equal, SE trials had longer estimated log mean RTs than ES trials ($p = .023$), with a difference of 0.043 ($SE = 0.019$, $p = .023$) in the condition of interest: feedback and short CSI. The interaction between trial type and CSI, $F(1, 32,201) = 12.51$, $p < .001$, reflects the fact that the difference in estimated mean log-transformed outcomes between switching and repetition trials is higher at short CSIs than at long CSIs (see Figure 3(a)).

CSI length was a very good predictor of performance on its own, $F(1, 32,201) = 492.52$, $p < .001$, with shorter estimated average log RTs occurring at longer CSIs than at short CSIs keeping all else constant. Furthermore, the presence of feedback was associated with longer estimated mean log RTs, $F(1, 32,201) = 28.99$, $p < .001$, and its effect depended on the CSI. The disadvantage brought by the presence of feedback was smaller at long CSIs than at short CSIs, $F(1, 32,201) = 32.50$, $p < .001$, and was influenced by trial type, $F(1, 32,201) = 13.20$, $p < .001$, being larger during ES trials than EE trials.

Effects of age and gender

There were also differences in performance related to age, $F(1, 104) = 36.64$, $p < .001$, and gender, $F(1, 104) = 4.38$, $p = .04$. A one-year difference in age between two participants was associated with an estimated difference of 0.084 on the log scale; in other words, the one-year older child is expected to have the geometric mean of RTs shorter by 8.4% than the younger child (all other things being equal).

Table 2. Models of response times for all trials (a) and emotion–emotion (EE) trials (b).

Predictor	Estimate	95% CI		Pr (> t)
		Lower	Upper	
(a) Main model (38,880 observations)				
Intercept	1672.375	1590.812	1754.607	<.001
Feedback: no feedback	0.917	0.888	0.995	<.001
CSI: long	0.856	0.845	0.868	<.001
Trial type: ES	1.097	1.070	1.125	<.001
Trial type: SE	1.122	1.093	1.150	<.001
Trial type: SS	1.019	0.993	1.046	.148
Gender: male	1.073	1.004	1.146	.039
Age (in months)	0.993	0.991	0.995	<.001
Internalizing	1.004	1.001	1.006	.009
Feedback × CSI	1.048	1.030	1.064	<.001
CSI × trial type ES	0.952	0.929	0.975	<.001
CSI × trial type SE	0.934	0.911	0.958	<.001
CSI × trial type SS	0.984	0.962	1.007	.168
Feedback × trial type ES	1.050	1.025	1.077	<.001
Feedback × trial type SE	1.065	1.039	1.093	<.001
Feedback × trial type SS	0.989	0.963	1.016	.433
Trial type ES × internalizing	0.999	0.998	1.000	.005
Trial type SE × internalizing	0.999	0.998	1.000	.092
Trial type SS × internalizing	0.999	0.998	1.000	.080
Feedback × internalizing	0.998	0.996	0.999	.045
Trial type ES × gender	0.970	0.943	0.990	.006
Trial type SE × gender	1.041	1.015	1.068	.002
Trial type SS × gender	0.970	0.940	0.992	.012
(b) Model for EE trials (19,224 observations)				
Intercept	1581.296	1505.680	1660.709	<.001
Feedback: no feedback	0.918	0.902	0.934	<.001
CSI: long	0.856	0.841	0.872	<.001
Gender: male	1.071	1.002	1.145	.043
Age (in months)	0.994	0.992	0.996	<.001
Internalizing	1.004	1.002	1.006	.004
Feedback × CSI	1.052	1.028	1.077	<.001
Feedback × internalizing	0.998	0.997	0.999	<.001
Valence: negative–neutral	1.129	1.092	1.167	<.001
Valence: negative–positive	1.040	1.018	1.062	<.001
Valence: neutral–negative	1.213	1.182	1.244	<.001
Valence: neutral–neutral	1.217	1.163	1.273	<.001
Valence: neutral–positive	1.171	1.142	1.201	<.001
Valence: positive–negative	1.069	0.090	1.044	<.001
Valence: positive–neutral	1.088	0.105	1.064	<.001
Valence: positive–positive	1.023	0.043	1.003	.025

Note: Baseline coding: EE trials, feedback present, short CSI and female gender. CSI = cue-to-stimulus interval; SS = shape–shape repetition; ES = emotion–shape switch; SE = shape–emotion switch.

Exponentiated parameter estimates for the models of log-transformed response times.

All covariates are mean centered.

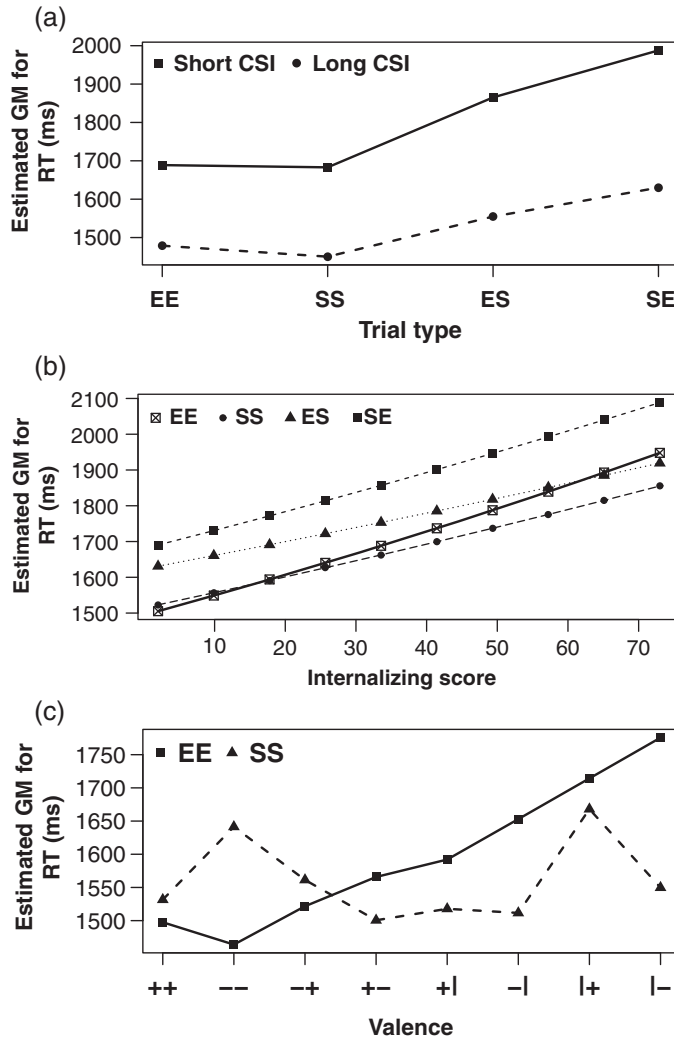


Figure 3. (a) Estimated geometric means of response time by trial type and cue-to-stimulus interval (CSI). (b) Estimated geometric means of response time by internalizing symptoms and trial type in the feedback condition. (c) Estimated geometric means of response time by valence for repetition trials. Note: Short CSI = 200 ms; long CSI = 1500 ms; EE = emotion–emotion repetition; SS = shape–shape repetition; ES = emotion–shape switch; SE = shape–emotion switch; ++ = positive face–positive face; -- = negative face–negative face; +- = negative face–positive face; +- = positive face–negative face; ++ = positive face–neutral face; -| = negative face–neutral face; |+ = neutral face–positive face; |- = neutral face–negative face.

Boys took longer to respond than girls, but the effect of gender depended on the trial type, $F(3, 32, 201) = 10.91, p < .001$. The effect was largest for SE trials, as boys were estimated to have an 11% increase in the geometric mean compared to girls ($p = .01$). Boys also performed worse in the EE condition compared to girls (7% increase, $p = .03$), but post hoc tests revealed they did not differ significantly from girls in the SS ($p = .30$) and ES ($p = .30$) conditions.

Effect of internalizing symptoms

Higher internalizing scores were associated with a general detrimental effect on RTs when controlling for age, gender, and the experimental manipulations, $F(1, 104) = 7.12$, $p = .01$. The effect of internalizing symptoms differed as a function of feedback presentation, $F(1, 32,201) = 4.03$, $p = .04$, and trial type, $F(1, 32,201) = 3.03$, $p = .03$, as suggested by Figure 3(b) which charts the estimated geometric means for RTs in each condition. Internalizing had a larger impact on trials where feedback was presented, and within this condition it was most pronounced for EE trials ($b = 0.004$, $SE = 0.001$, $p = .004$). Thus, an increase of 10 points on the internalizing scale was estimated to lead to a 1.04 times larger expected geometric mean for RTs. The effect of internalizing was non-significant in EE trials without feedback ($p = .31$), and overall non-significant for ES trials ($p = .42$), and then for SE ($p = .16$) and SS trials ($p = .24$) after the application of multiple testing corrections, although still in the range of interest for exploratory studies.

Post hoc analysis of stimulus valence

The stimuli in the experiment were valence-sensitive and switches occurred at times between stimuli with different emotional valences. We wanted to explore potential differences in performance related to the valence of the stimuli, and ultimately to adjust for valence as this could be a potential confounder of the effects of the other experimental manipulations. However, due to the presence of different types of valences with switching and repetition trials, as well as the small number of trials within each valence switch category, we only included this predictor in post hoc analyses.³ Valence was a significant predictor only for RTs of repetition trials. In order to better understand the effects of valence, two models were estimated for EE and SS trials. Again, accuracy trade-offs were overruled by a logistic model as in the previous section. The comparisons between the different valences did not lead to a facile interpretation in the case of SS trials (the estimated marginal geometric means of RTs for each valence can be observed in Figure 3(c)). However, in EE trials, valence was not only a highly significant predictor, $F(8, 1655) = 48.59$, $p < .001$, but it also appears that repetitions between two emotionally valenced stimuli were easier to complete than repetitions between neutral–emotional and emotional–neutral stimuli. The regression estimates for the fixed effects model are presented in Table 2b.

Discussion

The aim of the current study was to examine how emotional attention set-shifting links to individual differences in internalizing symptoms during an under-investigated developmental period (7–11 years). An initial objective was to identify a developmentally appropriate task to assess the ability to flexibly alternate between emotional and nonemotional judgments that would be sensitive to individual differences in internalizing symptoms. The ACCE task selected and adapted for this purpose represents an innovative experimental paradigm designed to assess both a mechanism of emotion regulation (i.e. attentional deployment) and individual differences in attentional control (Johnson, 2009a). To our knowledge, this is the first time that an affective task-switching paradigm was used during middle childhood (but see Wilson, Derryberry, & Kroeker, 2007, for an

investigation with a different affective task-switching paradigm in younger children). Our results attest the modified ACCE as a valid measure that can be used across various age groups to evaluate emotional attention set-shifting. The results replicated the common finding of the task-switching paradigm (e.g. Monsell, 2003): more errors and longer RTs in switching compared to repetition trials. Shorter CSIs (200 vs. 1500 ms) resulted in losses in performance effectiveness and efficiency, and this affected responses on switching trials to a larger degree than on repetition trials. Importantly, the results of both accuracy and RT measures converged on these findings, dismissing the possibility of a speed-accuracy trade-off. Replicating Johnson's (2009a) findings, we found longer RTs for disengaging a neutral set and engaging an emotional set than for the opposite switch (emotional to neutral).

From a developmental point of view, the predicted age-related improvement was most visible in terms of RTs, older children being significantly faster in their overall response speed. This confirms the constant rate of improvement in set-shifting response speed during middle childhood reported by previous studies (e.g. Cragg & Nation, 2009; Huizinga et al., 2006; Rosso, Young, Femia, & Yurgelun-Todd, 2004). This progress was translated into a modest age-related reduction in error rates, considering that overall accuracy on the ACCE task was already high even for the youngest children. We also noted an interesting gender-related difference in response speed, boys presenting longer RTs compared to girls in making an emotional judgment after performing a nonemotional, or a different emotional judgment. However, this difference was not present when switching to or when repeating a nonemotional judgment. Girls' advantage in engaging an emotional set could rely on a more general female advantage in facial emotion processing which has been reported during childhood and adolescence (McClure, 2000).

In line with the ACT, higher levels of anxiety (and depression) did not impact performance effectiveness but affected performance efficiency under certain circumstances. The negative association between internalizing symptoms and response speed was present only at the level of EE trials. Since EE trials are repetition trials (emotion following emotion judgments), this result was surprising at first. It is possible that internalizing symptoms influenced performance in EE trials due to the need for sustained manipulation of emotional sets during these trials. SE and ES trials also require the manipulation of emotional sets, but to a lower extent, thus the source of the difference could be quantitative rather than qualitative. It is tempting to speculate that the effect of internalizing symptoms has more to do with the sustained and flexible use of emotional sets rather than with the mental set switching ability in itself (at least during this developmental period).

However, the apparent lack of an internalizing-related switching impairment is arguable, since a degree of "emotional valence switching" was also involved in repetition trials as emotional valences varied between stimuli from the same mental set. It is possible that this "emotional valence switching" is more salient in EE trials since it matches the goal of the task (the cue indicated that the children should pay attention to the "emotional" content). Furthermore, the effect of internalizing symptoms on RTs was not moderated by the effect of valence switch type. This could lead to the conclusion that the effect of internalizing symptoms is not tied to a specific emotional valence switch (say, to or from stimuli with negative emotionality), or even if that is the case, it is only present under certain conditions (such as task-relevance).

A collateral aim was to investigate whether self-reported attentional control capacity could act as a protecting factor against the adverse effects of internalizing symptoms on emotional attention set-shifting. This hypothesis was not confirmed in the present study. A preliminary note explaining this finding relates to a discrepancy with previous findings (e.g. Muris, van der Pennen, Sigmond, & Mayer, 2008), namely the lack of a significant association between ACS-C scores and psychopathological (especially anxiety) symptoms. Moreover, there was also no age-related improvement on this self-report measure of attentional control. All these arguments call into doubt the capacity of the ACS-C as measured in our sample to adequately assess (self-perceived) attentional control. Moreover, the ACCE task is a behavioral measure of attentional control and previous studies indicated that self-report and performance-based measures of attention/effortful control are not strongly related (Muris et al., 2008).

Looking at the effects of valence, post hoc analyses revealed a significant effect on RTs only in the case of repetition trials. In EE trials particularly, consecutive emotional–emotional facial expression judgments were associated with shorter RTs than consecutive trials which included an emotional and a neutral facial expression. This is in line with the previous speculation about the existence of an “emotional switching” cost between emotional and neutral facial expressions. The pattern of results can be accounted for by an automatic exogenous process whereby attention is drawn preferentially to emotional stimuli (Vuilleumier, Armony, Driver, & Dola, 2001), which predictably aids a task that explicitly requires the repeated judgment of emotionality.

A final aim was to explore the role of feedback in relation to internalizing symptoms. When feedback was offered, participants adopted a more cautious response strategy, which translated into longer RTs. This strategy also resulted in fewer errors, which can suggest a trade-off between accuracy and speed in the feedback condition. The investigation of their interrelationships was incomplete since our analysis excluded trials that followed negative feedback. However, the fact that the mere presence of (positive) feedback was not facilitative, but rather detrimental to the performance efficiency is consequential in itself. Moreover, the detrimental effect of internalizing symptoms was noted only in the presence of (positive) feedback. Previous studies have only identified a relationship between negative feedback and exaggerated responses to perceived failure in children with higher levels of sadness/negative affect (Luciana, Gunnar, Davis, Nelson, & Donzella, 2005).

When discussing the effects of internalizing symptoms, we need to keep in mind that the anxiety levels of our sample were not very high such that results might be different when contrasting more extreme anxiety levels. Another limitation was that we did not measure arousal levels for the facial expressions, which is problematic because negative facial expressions have been shown to trigger higher arousal levels than those portraying positive emotions (Lang, Greenwald, Bradley, & Hamm, 1993). Moreover, studies on adults have dissociated the impact of emotional stimulus valence and arousal on task-switching performance (Demanet et al., 2011). Furthermore, the study did not include a measure of state anxiety, which might impact attention set-shifting independent of trait anxiety (Pacheco-Unguetti, Acosta, Callejas, & Lupianez, 2010; Visu-Petra, Miclea, & Visu-Petra, 2012). There are also some remaining concerns related to the design of the study, in particular, the fact that additional controls are needed to make sure that the effect of task switching is not confounded by the potential effect of “emotional switching.”

The present study brings several contributions to the very limited literature on emotional attentional control in children. First, it shows that during middle childhood children are already quite proficient in flexibly switching between emotional and nonemotional judgments according to contextual cues, although their processing efficiency continues to improve gradually. Girls appear to be faster than boys in engaging emotional sets. Across genders, higher levels of internalizing symptoms are negatively related to children's speed of alternating between two emotional judgments. Also, irrespective of internalizing levels, children have difficulties especially in making repeated emotional judgments related to consecutive neutral and emotional facial expressions. When trial-by-trial feedback is provided, children are more cautious and make fewer errors, and this tendency is stronger in children with higher levels of internalizing symptoms.

Further research within the individual differences framework needs to be carried out in order to determine the specific conditions under which emotional valence interacts with internalizing symptoms to modulate attention shifting during middle childhood. Cross-sectional, and especially longitudinal studies conducted earlier during development could clarify the reciprocal links between emotional attention set-shifting and the emergence or proliferation of internalizing symptoms during the transition from childhood to adolescence. Besides threat-related inhibition failures (Kindt & van der Hout, 2001), the difficulties experienced by children with internalizing symptoms in alternating between emotional judgments might be related to their less efficient emotional regulation skills (Eisenberg et al., 2001). Finally, additional investigations should be carried out to capture the critical conditions under which emotional valence has an influence on the ability to flexibly adapt to a permanently changing environment.

Acknowledgments

The authors are grateful to the children, parents, and school staff members who were involved in this project.

Funding

This work was supported by a grant from the Romanian National Authority for Scientific Research, CNCS – UEFISCDI, project number [PNII-ID-PCCE-2011-2-0045].

Notes

1. This is consistent with previous findings of informant discrepancies in the assessment of internalizing symptoms (see De Los Reyes & Kazdin, 2005, for a review).
2. The agreement between the parental and child internalizing scores (measured as the absolute difference between the parent and child score) did not correlate with the age of the children, $r(84) = -.11$, $p = .19$, two-tailed.
3. Keeping this limitation in mind, we note that the addition of valence to the main model did not change the direction or magnitude of the switching effect.

References

- Allport, A., Styles, E. A., & Hsieh, S. L. (1994). Shifting intentional set-exploring the dynamic control of tasks. *Attention and Performance XV: Conscious and Nonconscious Information Processing*, 15, 421–452.

- Ansari, T. L., Derakshan, N., & Richards, A. (2008). Effects of anxiety on task switching: Evidence from the mixed antisaccade task. *Cognitive Affective & Behavioral Neuroscience*, 8(3). doi:[10.3758/cabn.8.3.229](https://doi.org/10.3758/cabn.8.3.229)
- Bates, D., & Maechler, M. (2010). *lme4*: Linear mixed-effects models using Eigen and Eigen++ (Version 0.999375-37) [R package]. Retrieved from <http://CRAN.R-project.org/package=lme4> for R package version 0.999375-37.
- Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108, 624–652. doi:[10.1037/0033-295X.108.3.624](https://doi.org/10.1037/0033-295X.108.3.624)
- Chorpita, B. F., Yim, L., Moffitt, C., Umemoto, L. A., & Francis, S. E. (2000). Assessment of symptoms of DSM-IV anxiety and depression in children: A Revised Child Anxiety and Depression Scale. *Behaviour Research and Therapy*, 38, 835–855. doi:[10.1016/S0005-7967\(99\)00130-8](https://doi.org/10.1016/S0005-7967(99)00130-8)
- Cragg, L., & Nation, K. (2009). Shifting development in mid-childhood: The influence of between-task interference. *Developmental Psychology*, 45, 1465–1479. doi:[10.1037/a0015360](https://doi.org/10.1037/a0015360)
- Davidson, M. C., Amso, D., Cruess Anderson, L., & Diamond, A. (2006). Development of cognitive control and executive functions from 4 to 13 years: Evidence from manipulations of memory, inhibition, and task switching. *Neuropsychologia*, 44, 2037–2078. doi:[10.1016/j.neuropsychologia.2006.02.006](https://doi.org/10.1016/j.neuropsychologia.2006.02.006)
- De Lissnyder, E., Koster, E. H. W., Derakshan, N., & De Raedt, R. (2010). The association between depressive symptoms and executive control impairments in response to emotional and non-emotional information. *Cognition & Emotion*, 24, 264–280. doi:[10.1080/02699930903378354](https://doi.org/10.1080/02699930903378354)
- De Los Reyes, A., & Kazdin, A. E. (2005). Informant discrepancies in the assessment of childhood psychopathology: A critical review, theoretical framework, and recommendations for further study. *Psychological Bulletin*, 131, 483–509. doi:[10.1037/0033-2909.131.4.483](https://doi.org/10.1037/0033-2909.131.4.483)
- Demane, J., Liefvooghe, B., & Verbruggen, T. A. (2011). Valence, arousal, and cognitive control: A voluntary task-switching study. *Frontiers in Psychology*, 2, 336. doi:[10.3389/fpsyg.2011.00336](https://doi.org/10.3389/fpsyg.2011.00336)
- Derakshan, N., & Eysenck, M. W. (2009). Anxiety, processing efficiency, and cognitive performance: New developments from attentional control theory. *European Psychologist*, 14, 168–176. doi:[10.1027/1016-9040.14.2.168](https://doi.org/10.1027/1016-9040.14.2.168)
- Derakshan, N., Smyth, S., & Eysenck, M. W. (2009). Effects of state anxiety on performance using a task-switching paradigm: An investigation of attentional control theory. *Psychonomic Bulletin & Review*, 16, 1112–1117. doi:[10.3758/PBR.16.6.1112](https://doi.org/10.3758/PBR.16.6.1112)
- Derryberry, D., & Reed, M. A. (2002). Anxiety-related attentional biases and their regulation by attentional control. *Journal of Abnormal Psychology*, 111, 225–236. doi:[10.1037/0021-843X.111.2.225](https://doi.org/10.1037/0021-843X.111.2.225)
- Eisenberg, N., Cumberland, A., Spinrad, T. L., Fabes, R. A., Shepard, S. A., Reiser, M., & Guthrie, I. K. (2001). The relations of regulation and emotionality to children's externalizing and internalizing problem behavior. *Child Development*, 72, 1112–1134. doi:[10.1111/1467-8624.00337](https://doi.org/10.1111/1467-8624.00337)
- Emerson, C. S., Mollet, G. A., & Harrison, D. W. (2005). Anxious-depression in boys: An evaluation of executive functioning. *Archives of Clinical Neuropsychology*, 20, 539–546. doi:[10.1016/j.acn.2004.10.003](https://doi.org/10.1016/j.acn.2004.10.003)
- Eysenck, M. W., Derakshan, N., Santos, R., & Calvo, M. G. (2007). Anxiety and cognitive performance: Attentional control theory. *Emotion*, 7, 336–353. doi:[10.1037/1528-3542.7.2.336](https://doi.org/10.1037/1528-3542.7.2.336)
- Field, A. P., & Wright, D. B. (2011). A primer on using multilevel models in clinical and experimental psychopathology research. *Journal of Experimental Psychopathology*, 2, 271–293. doi:[10.5127/jep.013711](https://doi.org/10.5127/jep.013711)
- Fox, J. (2003). Effect displays in R for generalised linear models. *Journal of Statistical Software*, 8(15), 1–27.
- Heim, S., & Keil, A. (2012). Developmental trajectories of regulating attentional selection over time. *Frontiers in Psychology*, 3, 277. doi:[10.3389/fpsyg.2012.00277](https://doi.org/10.3389/fpsyg.2012.00277)
- Hothorn, T., Bretz, F., & Westfall, P. (2008). Simultaneous inference in general parametric models. *Biometrical Journal*, 50, 346–363. doi:[10.1002/bimj.200810425](https://doi.org/10.1002/bimj.200810425)
- Huizinga, M., Dolan, C. V., & van der Molen, M. W. (2006). Age-related change in executive function: Developmental trends and a latent variable analysis. *Neuropsychologia*, 44, 2017–2036. doi:[10.1016/j.neuropsychologia.2006.01.010](https://doi.org/10.1016/j.neuropsychologia.2006.01.010)

- Johnson, D. R. (2009a). Attentional control capacity for emotion: An individual-difference measure of internal controlled attention. *Cognition & Emotion*, 23, 1516–1536. doi:10.1080/02699930802437095
- Johnson, D. R. (2009b). Emotional attention set-shifting and its relationship to anxiety and emotion regulation. *Emotion*, 9, 681–690. doi:10.1037/a0017095
- Kiesel, A., Steinhauser, M., Wendt, M., Falkenstein, M., Jost, K., Philipp, A. M., & Koch, I. (2010). Control and interference in task switching – A review. *Psychological Bulletin*, 136, 849–874. doi:10.1037/a0019842
- Kindt, M., & van den Hout, M. (2001). Selective attention and anxiety: A perspective on developmental issues and the causal status. *Journal of Psychopathology and Behavioral Assessment*, 23(3), 193–202. doi:10.1023/A:1010921405496
- Lang, P. J., Greenwald, M. K., Bradley, M. M., & Hamm, A. O. (1993). Looking at pictures: Affective, facial, visceral, and behavioral reactions. *Psychophysiology*, 30, 261–273. doi:10.1111/j.1469-8986.1993.tb03352.x
- Lonigan, C. J., Carey, M. P., & Finch, A. J. (1994). Anxiety and depression in children and adolescents: Negative affectivity and the utility of self-reports. *Journal of Consulting and Clinical Psychology*, 62, 1000–1008. doi:10.1037/0022-006X.62.5.1000
- Luciana, M., Gunnar, M. R., Davis, E. P., Nelson, C. A., & Donzella, B. (2005). Children's "catastrophic responses" to negative feedback on CANTAB's ID/ED set-shifting task: Relation to indices of a depressive temperamental style. *Cognition, Brain, Behavior*, 9, 343–361.
- McClure, E. B. (2000). A meta-analytic review of sex differences in facial expression processing and their development in infants, children, and adolescents. *Psychological Bulletin*, 126, 424–453. doi:10.1037/0033-2909.126.3.424
- Monsell, S. (1996). Control of mental processes. In V. Bruce (Ed.), *Unsolved mysteries of the mind: Tutorial essays in cognition* (pp. 93–148). Hove: Lawrence Erlbaum.
- Monsell, S. (2003). Task switching. *Trends in Cognitive Sciences*, 7, 134–140. doi:10.1016/S1364-6613(03)00028-7
- Muris, P., van der Pennen, E., Sigmond, R., & Mayer, B. (2008). Symptoms of anxiety, depression, and aggression in non-clinical children: Relationships with self report and performance-based measures of attention and effortful control. *Child Psychiatry and Human Development*, 39, 455–467. doi:10.1007/s10578-008-0101-1
- Pacheco-Unguetti, A. P., Acosta, A., Callejas, A., & Lupianez, J. (2010). Attention and anxiety: Different attentional functioning under state and trait anxiety. *Psychological Science*, 21, 298–304. doi:10.1177/0956797609359624
- Perez-Edgar, K., & Fox, N. A. (2007). Temperamental contributions to children's performance in an emotion-word processing task: A behavioral and electrophysiological study. *Brain and Cognition*, 65(1), 22–35. doi:10.1016/j.bandc.2006.10.010
- Pinheiro, J., Bates, D., DebRoy, S., Sarkar, S., & R Development Core Team. (2010). *nlme: Linear and nonlinear mixed effects models* (Version 3.1-97) [R package version]. Retrieved from <http://CRAN.R-project.org/package=lme4> for R package version 3.1-97.
- Rogers, M. A., Kasai, K., Koji, M., Fukuda, R., Iwanami, A., Nakagome, K., ... Kato, N. (2004). Executive and prefrontal dysfunction in unipolar depression: A review of neuropsychological and imaging evidence. *Neuroscience Research*, 50(1), 1–11. doi:10.1016/j.neures.2004.05.003
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124, 207–231. doi:10.1037/0096-3445.124.2.207
- Rosso, I. M., Young, A. D., Femia, L. A., & Yurgelun-Todd, D. A. (2004). Cognitive and emotional components of frontal lobe functioning in childhood and adolescence. *Annals of the New York Academy of Sciences*, 1021, 355–362. doi:10.1196/annals.1308.045
- Sibrava, N. J., & Borkovec, T. D. (2006). The cognitive avoidance theory of worry. In G. C. L. Davey & A. Wells (Eds.), *Worry and its psychological disorders: Theory, assessment, and treatment* (pp. 239–258). Chichester: Wiley.
- Spence, S. H. (1997). Structure of anxiety symptoms among children: A confirmatory factor-analytic study. *Journal of Abnormal Psychology*, 106, 280–297. doi:10.1037/0021-843X.106.2.280
- Spence, S. H. (1998). A measure of anxiety symptoms among children. *Behaviour Research and Therapy*, 36, 545–566. doi:10.1016/S0005-7967(98)00034-5

- Susa, G., Pitica, I., Benga, O., & Miclea, M. (2012). The self regulatory effect of attentional control in modulating the relationship between attentional biases toward threat and anxiety symptoms in children. *Cognition & Emotion*, 26, 1069–1083. doi:[10.1080/02699931.2011.638910](https://doi.org/10.1080/02699931.2011.638910)
- Tincas, I., Cheie, L., Mocan, O., Benga, O., & Visu-Petra, L. (in preparation). Validating the Revised Child Anxiety and Depression Scale on the Romanian population.
- Tincas, I., Dragos R., Ionescu, T., & Benga, O. (2007). Attentional set-shifting in preschoolers: Anxiety-related response patterns. *Cognition, Brain, Behavior*, 3, 553–570.
- Tottenham, N., Borscheid, A., Ellertsen, K., Marcus, D. J., & Nelson, C. A. (2002). Categorization of facial expressions in children and adults: Establishing a larger stimulus set. *Journal of Cognitive Neuroscience*, 14, S74.
- Visu-Petra, L., Miclea, M., & Visu-Petra, G. (2012). Individual differences in anxiety and executive functioning: A multidimensional view. *International Journal of Psychology*, 48(4), 1–11. doi:[10.1080/00207594.2012.656132](https://doi.org/10.1080/00207594.2012.656132)
- Vuilleumier, P., Armony, J. L., Driver, J., & Dola, R. J. (2001). Effects of attention and emotion on face processing in the human brain: An event-related fMRI study. *Neuron*, 30, 829–841. doi:[10.1016/S0896-6273\(01\)00328-2](https://doi.org/10.1016/S0896-6273(01)00328-2)
- Wilson, B. J., Derryberry, D., & Kroeker, D. (2007). A computerized task for investigating the relation between attentional and emotional processes in children. *The Journal of Genetic Psychology*, 167, 415–431. doi:[10.3200/GNTP.167.4.415-432](https://doi.org/10.3200/GNTP.167.4.415-432)

Copyright of Anxiety, Stress & Coping is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.